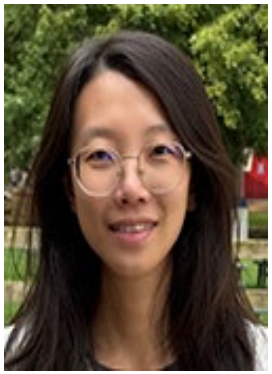
Measurement Error Models

- Donna Spiegelman
- Molin Wang
- Jingyu Cui
- Raymond Carroll

Who We Are

Donna

Molin



Jingyu

Raymond

Speakers

- Donna Spiegelman, Yale SPH, formerly Harvard SPH, over 800 papers, over 40 years in the field
- Molin Wang, Harvard SPH, Epi, Biostat, many papers, h-index 71, i10-index 220
- Jingyu Cui, Yale SPH, Postdoc with Donna
- Raymond J. Carroll, TAMU, Stat
- Good, brief biosketches via AI, e.g., coPilot

Outline

- Introduction (Carroll, Spiegelman, 0.5 hr)
- Validation and replication study designs (Spiegelman, 0.5 hr)
- Regression calibration (Wang, 1.0 hr)
- Lab examples using R (Cui, 0.75 hr, then 15 minute break)
- Integration with machine learning (Wang, 0.5 hr)
- Examples in nutrition and environment (Spiegelman, 0.5 hr)

Introduction

Outline

- Measurement Error Examples
- Some general results

The Triple Whammy of Measurement Error

- **Bias** in parameter estimation
- **Loss of Power** for detecting signals
- **Masking** the features of the data: measurement error causes the model of the observed data to change

Data

- $D =$ binary, $Y =$continuous, $T =$ survival
- $X =$ true "exposure"
- $Z =$ surrogate for X, e.g., $Z = X + \widetilde{\epsilon}$
- $Z = X + \epsilon$ or $Z = E(X|Z) + \widetilde{\epsilon}$
- $V =$ Other error-free variables
- Example: $Y = \beta_0 + \beta_1 X + \beta_v V + e$

Simple Approach

- Better to replace Z by a function of (Z, V), namely $E(X|Z, V)$, since $X = E(X|Z, V) + \widetilde{\epsilon}$

Why Special Methods Are Needed — Cont.

- **New Model**: $E(Y|Z,V) = \beta_0 + \beta_1 E(X|Z,V) + \beta_v V$ so

$$[Y|Z,V] = \beta_0 + \beta_1 E(X|Z,V) + \beta_v V + (e + \beta_1 \widetilde{\epsilon})$$

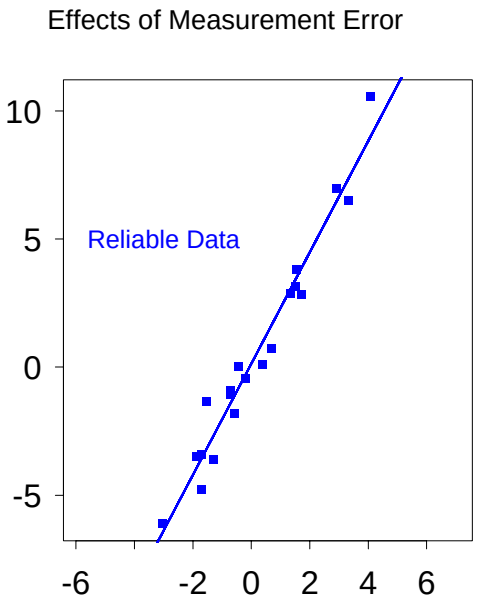
  - Special case of "**regression calibration**"
  - Leads to unbiased estimates
  - Notice the inflation of variance: $\sigma_e^2 + \beta_1^2 \sigma_{\widetilde{\epsilon}}^2$.

Examples Of Measurement Error Problems

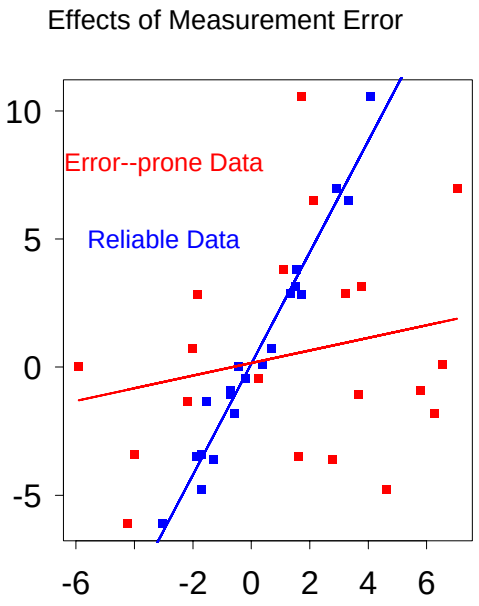
- Measuring usual nutrient intake
- Measuring Systolic Blood Pressure
- Measuring Radiation Dose from the Nevada Test Site
- Measuring Exposure to arsenic in drinking water, dust in the workplace, radon gas in the home, and other environmental hazards
- **Others, such as survival**

Simulation Study

- Generate $X_1, \ldots, X_{50}$, IID $\mathsf{N}(0, 1)$
- Generate $Y_i = \beta_0 + \beta_1 X_i + e_i$
  - e_i IID $\mathsf{N}(0, 1/9)$
  - $\beta_0 = 0$
  - $\beta_1 = 1$
- Generate $\widetilde{\epsilon}_1, \ldots, \widetilde{\epsilon}_{50}$, IID $\mathsf{N}(0, 1)$
- Set $Z_i = X_i + \widetilde{\epsilon}_i$
- Regress Y on X and Y on Z and compare



True Data Without Measurement Error



Observed Data With Measurement Error

Multiple Linear Regression

- **Triple Whammy**
  - **Bias**
  - **Increased variance** and loss of power
  - **Loss of features**

- In multiple linear regression, the bias can take unusual forms

- This can lead to **invalidity of hypothesis testing**

Multiple Linear Regression With Error

- **Model**

$$
\begin{aligned}
Y &= \beta_0 + \beta_1^{\mathsf{T}} X + \beta_v^{\mathsf{T}} V + e \\
Z &= X + \widetilde{\epsilon}
\end{aligned}
$$

- **Bias** Regressing Y on Z and V estimates

$$
\begin{pmatrix} \beta_{1*} \\ \beta_{v*} \end{pmatrix} = \Lambda \begin{pmatrix} \beta_1 \\ \beta_v \end{pmatrix} \qquad \left[\neq \begin{pmatrix} \beta_1 \\ \beta_v \end{pmatrix} \right]
$$

- For Λ, see next

Multiple Linear Regression With Error, cont.

- Λ is the **attenuation matrix** or reliability matrix

$$\Lambda = \begin{pmatrix} \sigma_{xx} + \sigma_{\widetilde{\epsilon}\widetilde{\epsilon}} & \sigma_{xv} \\ \sigma_{vx} & \sigma_{vv} \end{pmatrix}^{-1} \begin{pmatrix} \sigma_{xx} & \sigma_{xv} \\ \sigma_{vx} & \sigma_{vv} \end{pmatrix}$$

- The terms σ_{xx}, etc. are now covariance matrices
- Biases can take many forms, including **reversal of signs**!
- **Global null test is OK**: Naive test of $H_0 : \beta_1 = 0$ and $\beta_v = 0$ is valid
- **No effect for any effect of X is OK**: Naive test of $H_0 : \beta_1 = 0$ is valid

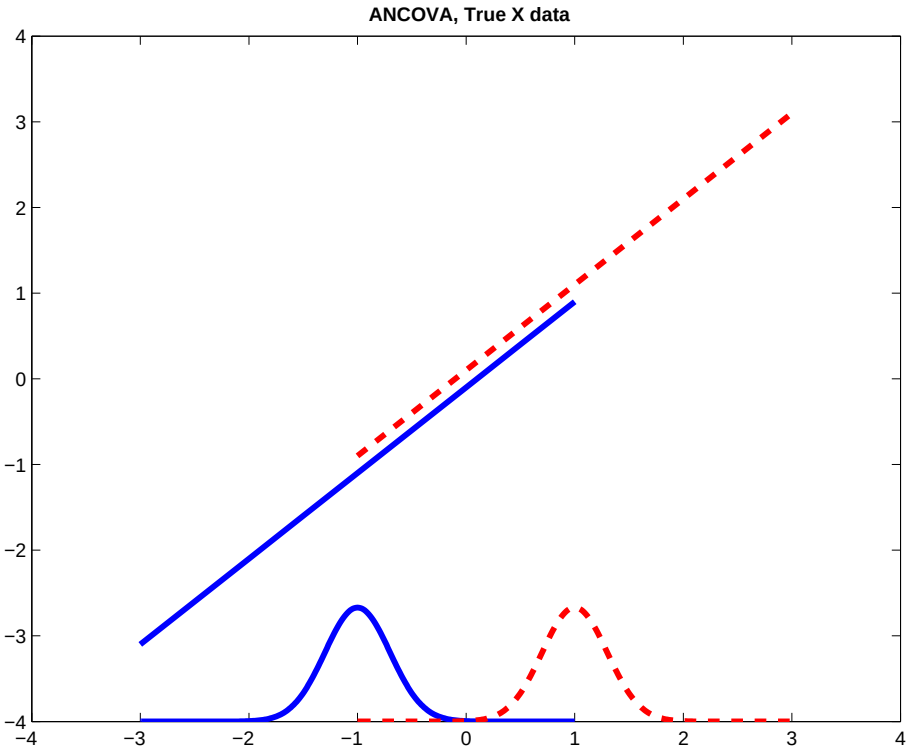
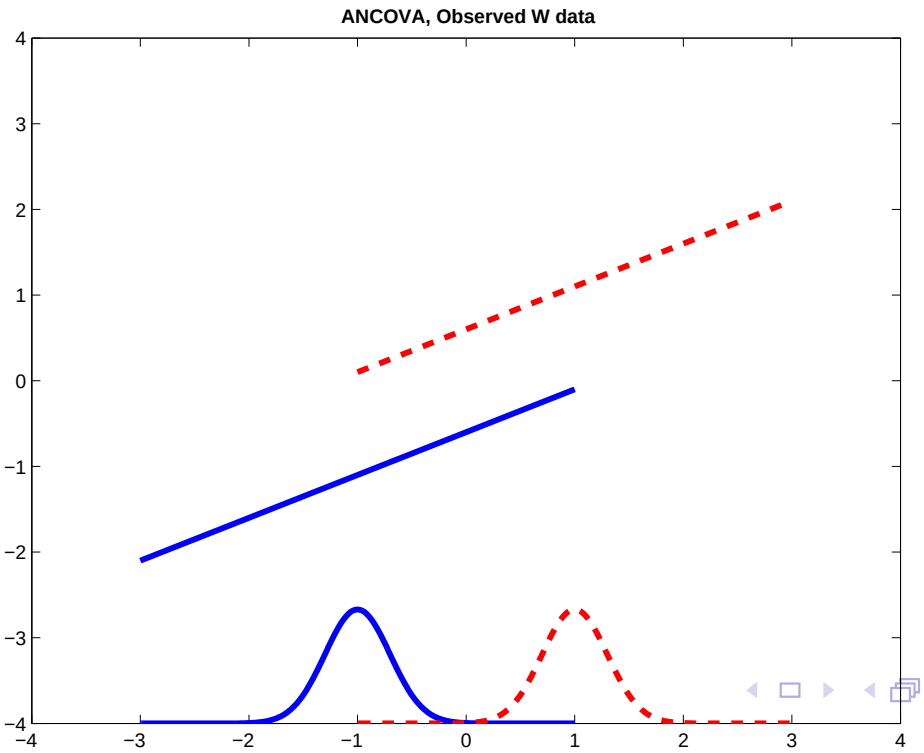
What Naive Tests are Invalid?

- **Tests for Z**: surprisingly, tests about the variables Z measured **without error** are **not always valid**
  - Exception is when X and Z are independent
- **When X is multivariate**: Naive tests here for single components of X are **not always valid**

Analysis Of Covariance

- These results have implications for the **two group ANCOVA**.
- **True Predictor** X
- **Group Assignment** V = dummy indicator of group 1, say
- **Imbalance**: If X has a different mean for the two groups, then the estimated effect of V is **biased**
- **Illustration**: The next slides illustrate that even when there is no V effect in truth, the observed data may indicate, **falsely**, that there is an apparent effect

2-Group ANCOVA, True X Data. Note no effect.



2-Group ANCOVA, Observed Z Data. Note apparent effect. Typo in header.